

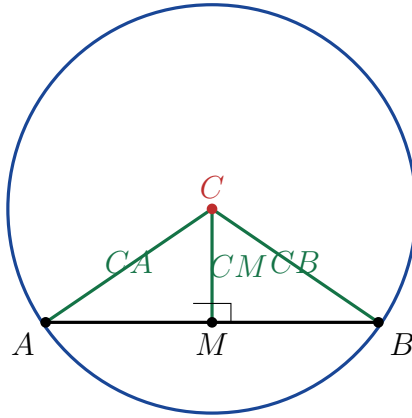
Question 1.

Can you explain why the converse to Theorem 4 is true, i.e., why does the perpendicular from the centre of a circle to a chord of the circle bisect the chord?

Hint: Use Fig. 5.12. You are told that $\angle CMA = \angle CMB = 90^\circ$. You need to show that $AM = BM$.

Solution

Let AB be a chord of a circle with centre C . The perpendicular from C meets the chord AB at M .



Given:

AB is a chord of the circle with centre C and

$$CM \perp AB.$$

Therefore,

$$\angle CMA = \angle CMB = 90^\circ.$$

To prove:

$$AM = BM.$$

Proof:

Consider $\triangle CMA$ and $\triangle CMB$.

Since CA and CB are radii of the same circle,

$$CA = CB.$$

Also,

$$CM = CM$$

as the side CM is common to both triangles.

Further,

$$\angle CMA = \angle CMB = 90^\circ.$$

Therefore,

$$\triangle CMA \cong \triangle CMB$$

by the **RHS congruence criterion**.

Hence, corresponding sides are equal.

Therefore,

$$\boxed{AM = BM}.$$

Thus, the perpendicular from the centre of a circle to a chord bisects the chord.

Hence proved.

Question 2.

An isosceles triangle ABC is inscribed in a circle, with $AB = AC$. Show that the altitude from A to BC passes through the centre of the circle.

Solution

Let AD be the altitude from A to BC .

Thus,

$$AD \perp BC.$$

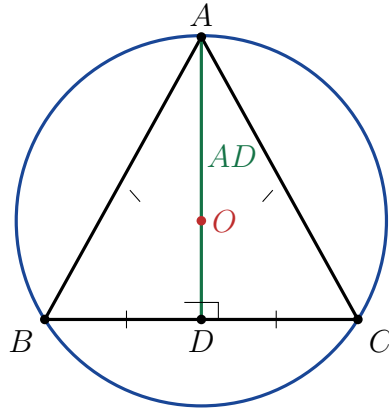
Since ABC is an isosceles triangle and

$$AB = AC,$$

the altitude from the vertex A to the base BC also bisects the base.

Therefore,

$$BD = DC.$$



Given:

$\triangle ABC$ is an isosceles triangle inscribed in a circle, where

$$AB = AC.$$

AD is the altitude from A to BC .

To prove:

The altitude AD passes through the centre O of the circle.

Proof:

Since

$$AB = AC,$$

$\triangle ABC$ is isosceles.

Therefore, the altitude from the vertex A to the base BC also bisects BC .

Hence,

$$BD = DC.$$

Also,

$$AD \perp BC.$$

Thus, AD is the perpendicular bisector of the chord BC .

We know that the perpendicular bisector of a chord passes through the centre of the circle.

Therefore, the centre O of the circle lies on AD .

Hence,

A, O, D are collinear.

Therefore, the altitude from A to BC passes through the centre of the circle.

Hence proved.

Question 3.

Two parallel chords of lengths 6 cm and 8 cm are on opposite sides of the centre of a circle. If the radius of the circle is 5 cm, find the distance between the midpoints of the chords.

Solution

Let the two parallel chords be AB and CD .

Given,

$$AB = 8 \text{ cm}, \quad CD = 6 \text{ cm}$$

and radius of the circle,

$$r = 5 \text{ cm}.$$

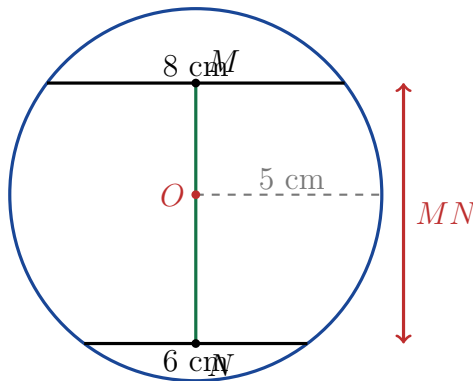
Let M and N be the midpoints of AB and CD respectively.

Since the perpendicular from the centre to a chord bisects the chord,

$$AM = MB = \frac{8}{2} = 4 \text{ cm}$$

and

$$CN = ND = \frac{6}{2} = 3 \text{ cm}.$$



Let OM be the perpendicular distance from the centre to the 8 cm chord.

In right-angled triangle OMA ,

$$OA = 5 \text{ cm}, \quad AM = 4 \text{ cm}.$$

By Pythagoras theorem,

$$OA^2 = OM^2 + AM^2.$$

Therefore,

$$5^2 = OM^2 + 4^2,$$

$$25 = OM^2 + 16,$$

$$OM^2 = 9,$$

$$\boxed{OM = 3 \text{ cm}}.$$

Now consider the 6 cm chord.

In right-angled triangle ONC ,

$$OC = 5 \text{ cm}, \quad CN = 3 \text{ cm}.$$

Again, by Pythagoras theorem,

$$OC^2 = ON^2 + CN^2.$$

Therefore,

$$5^2 = ON^2 + 3^2,$$

$$25 = ON^2 + 9,$$

$$ON^2 = 16,$$

$$\boxed{ON = 4 \text{ cm}}.$$

The two chords are on opposite sides of the centre O . Therefore, the distance between their midpoints is

$$MN = OM + ON.$$

Hence,

$$MN = 3 + 4$$

$$\boxed{MN = 7 \text{ cm}}.$$

Answer: $\boxed{7 \text{ cm}}$

IMPORTANT RESULTS

Perpendicular from centre to a chord bisects the chord

Perpendicular bisector of a chord passes through the centre

Altitude of an isosceles triangle bisects its base

For Q3, distance between midpoints = 7 cm